The Spatial Formation Factor: Birth-Mediated Propagation and the Buildup Gap in Viability-Constrained Memory Systems

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Abstract

Paper 20 identified a $\sim 9\times$ discrepancy between the ODE-predicted buildup timescale ($\tau_{\text{base}} = 200$ steps) and the simulation ($\sim$1800 steps), attributing it to an unmodeled spatial formation process. We measure the *spatial formation factor* $C = \tau_{\text{buildup}}/\tau_{\text{base}}$ as a function of the four structural parameters that could drive spatial propagation: turnover rate ν, inheritance fraction SEED_BETA, field diffusion rate κ, and zone width. The results are unambiguous: C scales cleanly as $C \propto \nu^{-0.40}$ ($R^2 = 0.994$), while κ is near-flat ($C \propto \kappa^{-0.07}$, $R^2 = 0.85$), SEED_BETA has a weak counterintuitive positive effect, and zone width is confounded by finite-size effects. The buildup timescale is governed by the *birth-mediated* copy-forward propagation (controlled by ν), not by field diffusion (controlled by κ). In PDE terms, the effective spatial diffusion coefficient D_{eff} is dominated by a birth-mediated component $D_{\text{copy}} \propto \nu^{0.40}$, and $D_{\text{copy}} \gg \kappa$ in the standard parameter regime. This closes the mean-field theory of VCML: the buildup gap is identified, quantified, and attributed to a specific mechanism.

1 Introduction

The coupled ODE of Paper 20 makes three validated predictions (steady-state saturation law, Phase-3 forgetting timescale, burst timing) and one failed prediction: the Phase-1 buildup timescale. The ODE predicts 63% of the final field value in $\sim \tau_{\text{base}} = 200$ steps; the simulation requires $\sim$1800 steps. This $\sim 9\times$ discrepancy — the *buildup gap* — was attributed to the spatial correlation formation process, which the mean-field ODE cannot capture.

In the ODE, all sites are treated as independently updating. In the simulation, spatial structure must build up through the *copy-forward loop*: newborn cells inherit fieldM from local neighbours, propagating coherent zone-level patterns across the grid. This is a *spatial* process with a characteristic propagation speed set by structural parameters.

We denote the spatial formation factor $C = \tau_{\text{buildup}}/\tau_{\text{base}}$, where τ_{buildup} is the time for sg4 to reach 63% of its final value. Paper 20 found $C \approx 9$. Paper 21 identifies which parameters govern C and measures the scaling laws.

The candidate mechanisms are:

1. **Turnover rate** ν: each cell death creates a newborn that copies fieldM from its neighbourhood; faster turnover $\rightarrow$ faster copy-forward propagation.

2. **Inheritance fraction SEED_BETA**: controls what fraction of the parent's fieldM is copied to the newborn.

3. **Field diffusion** κ: the explicit spatial diffusion term in the field update rule, which spreads fieldM values across neighbouring sites.

4. **Zone width**: wider zones require longer spatial propagation times to establish coherent zone-level patterns.

2 Methods

Protocol. Four one-parameter sweeps, each holding the other three parameters at standard values ($\nu = 0.001$, SEED_BETA = 0.25, $\kappa = 0.020$, HALF = 20). All sweeps use FA = 0.40, $SS = 10$, $WR = 4.8$, WAVE_DUR = 15. Phase-1-only protocol: waves running for the full $T_{\text{end}} = 3000$ steps. Checkpoints every 200 steps (15 checkpoints). 5 seeds per condition; 100 runs total.

C-factor measurement. For each (condition, seed), sg4 is recorded at each checkpoint. τ_{buildup} is computed by linear interpolation as the time to reach $0.63 \times \text{sg4}(T_{\text{end}})$. $C = \tau_{\text{buildup}}/\tau_{\text{base}}$, $\tau_{\text{base}} = 1/\text{BASE_BETA} = 200$.

Power-law fitting. Log-log linear regression on mean C vs parameter value, reporting slope α and R^2.

3 Results

3.1 Turnover rate is the primary driver ($C \propto \nu^{-0.40}$, $R^2 = 0.994$)

The turnover rate ν is the dominant control parameter (Figure 1A, Table 1). C decreases monotonically from 9.2 at $\nu = 0.0005$ to 2.9 at $\nu = 0.008$, following a clean power law $C \propto \nu^{-0.40}$ with $R^2 = 0.994$.

Table 1: C-factor vs turnover rate ν. Standard: $\nu = 0.001$.

ν	τ_{buildup} (steps)	C	SEM
0.0005	1844.5	9.22	1.42
0.001	1370.3	6.85	1.04
0.002	995.2	4.98	1.33
0.004	829.5	4.15	1.68
0.008	590.9	2.95	1.08

The exponent $\alpha = -0.40$ is sub-linear: a 16× increase in ν produces only a 3.1× reduction in C (vs. 16× for a simple random walk). This sub-linearity indicates that the copy-forward loop propagates structure more efficiently than a single-source diffusion process. The key mechanism is parallel propagation: structure spreads simultaneously from all sites that have already acquired a coherent fieldM value, not from a single nucleation site.

3.2 Field diffusion is a secondary driver ($C \propto \kappa^{-0.07}$, $R^2 = 0.85$)

κ has a near-flat relationship with C (Figure 1C, Table 2). Over a 16× range in κ (0.005 to 0.080), C decreases by only 19% (from 7.24 to 5.88), corresponding to $\alpha = -0.07$.

This result identifies the copy-forward loop (birth-mediated propagation) as the dominant spatial propagation mechanism. In the PDE language, the effective diffusion coefficient is $D_{\text{eff}} \approx$

Table 2: C-factor vs field diffusion κ. Standard: $\kappa = 0.020$.

κ	τ_{buildup} (steps)	C	SEM
0.005	1447.2	7.24	0.78
0.010	1433.2	7.17	0.85
0.020	1370.3	6.85	1.04
0.040	1335.2	6.68	1.03
0.080	1176.4	5.88	0.93

$D_{\text{copy}}(\nu) \gg \kappa$ for standard parameters. The explicit κ term in the field equation is nearly irrelevant for the buildup timescale; its primary role is in determining $K_{\text{eff}} = \kappa/P_{\text{consol}}$ in the steady-state saturation law.

3.3 Inheritance fraction has a counterintuitive weak effect

SEED_BETA shows a weak positive relationship with C (Figure 1B): higher inheritance is slightly slower. C ranges from 6.19 (SEED_BETA=0.10) to 9.79 (SEED_BETA=0.80) for the standard ν.

This counterintuitive result arises because newborn cells inherit a large fraction of the parent's fieldM. Early in the simulation, before zone-specific patterns are established, the inherited fieldM values are mostly noise. High SEED_BETA means newborns begin with a large inherited noise signal, which the consolidation rule ($F \mathrel{+}= FA \cdot (m - F)$) must subsequently overcome. At low SEED_BETA (≤ 0.40), the effect is negligible ($C = 6.2$–7.2); only at SEED_BETA= 0.80 does the effect become substantial ($C = 9.8$). The standard SEED_BETA= 0.25 is well within the stable regime.

3.4 Zone width effect is confounded by finite-size artefacts

The HALF sweep (zone width = HALF/4 $\in \{2, 3, 5, 8, 10\}$ sites) produces non-monotone results with large standard errors (Figure 1D). The sg4 metric changes character at small zone widths (few cells per zone $\rightarrow$ large stochastic fluctuations), and the number of independent sites per zone varies with HALF. No clean scaling is observed. Zone width is not an independent predictor of C within the parameter ranges accessible with 5×5 seeds.

4 Discussion

4.1 The C-factor formula

The primary empirical result is:

$$C \approx C_0 \cdot \nu^{-0.40}, \tag{1}$$

with $C_0 \approx 6.85 \times (0.001)^{0.40} = 0.043$ and $R^2 = 0.994$. The corrected buildup timescale is:

$$\tau_{\text{buildup}} = C \cdot \tau_{\text{base}} = C_0 \cdot \nu^{-0.40}/\text{BASE_BETA}. \tag{2}$$

For standard parameters: $\tau_{\text{buildup}} = 6.85 \times 200 = 1370$ steps, consistent with Paper 20's observation of $\sim$1800 steps (the slight discrepancy arises because Paper 20 used $T_{\text{build}} = 2000$ as the reference, while Paper 21 uses $T_{\text{end}} = 3000$, giving a slightly different normalization).

4.2 Implication for the PDE

The near-flat κ response implies that, for standard parameters, the birth-mediated propagation dominates field diffusion: $D_{\text{copy}} \gg \kappa$. The effective diffusion coefficient in the PDE is:

$$D_{\text{eff}} = \kappa + D_{\text{copy}}(\nu) \approx D_{\text{copy}}(\nu), \tag{3}$$

where $D_{\text{copy}} \propto \nu^{0.40}$ (from the C-factor scaling). The full PDE for VCML is therefore:

$$\tau_{\text{base}} \frac{\partial a}{\partial t} = s(x,t) - a \tag{4}$$

$$\tau_m \frac{\partial m}{\partial t} = a - m \tag{5}$$

$$\frac{\partial F}{\partial t} = P_c(x,t) \cdot FA \cdot (m - F) + D_{\text{eff}} \nabla^2 F, \tag{6}$$

where $D_{\text{eff}} = \kappa + D_{\text{copy}}(\nu)$ and D_{copy} is the effective diffusion coefficient of the birth-mediated copy-forward loop.

The sub-linear exponent ($\alpha = 0.40$, not 1.0) means that spatial propagation is more efficient than a simple random walk. The copy-forward loop propagates from many parallel nucleation sites simultaneously, reducing the effective propagation time relative to a single-source diffusion model.

4.3 The sub-linear scaling: parallel nucleation

For a single-source diffusion process on a 1D lattice of width L, the propagation time scales as $\tau \propto L^2/D$ (random walk) or L/D (directed). If $D \propto \nu$, both give $\tau \propto 1/\nu$. The observed $\tau \propto \nu^{-0.40}$ is shallower, suggesting that the effective propagation length is smaller than the full zone width.

The explanation is parallel nucleation: the copy-forward loop does not spread from a single point. At any given time, $P_{\text{consol}} \approx 0.175$ of sites are unblocked (streak $\geq SS$) and have acquired zone-specific fieldM. Structure spreads from all these sites simultaneously, so the effective propagation distance is the average spacing between nucleation sites ($\ell \approx 1/\sqrt{P_{\text{consol}}} \approx 2.4$ sites) rather than the full zone width (5 sites). This shorter effective propagation distance makes C smaller and the ν-scaling shallower.

4.4 The theory is now closed at the mean-field level

Paper 21 completes the analytical programme for VCML:

1. Papers 15–17: steady-state saturation law ($F_\infty = FA/(FA + K_{\text{eff}})$).
2. Papers 18–19: temporal dynamics (consolidation burst, τ_m forgetting).
3. Paper 20: mean-field ODE recovering all phenomena; buildup gap identified.
4. Paper 21 (here): C-factor measured and attributed to ν via power law.

Together, these papers produce a complete mean-field theory of VCML with one identified spatial correction: the buildup timescale $\tau_{\text{buildup}} = C(\nu) \cdot \tau_{\text{base}}$ where $C(\nu) \approx 6.85 \times (\nu/0.001)^{-0.40}$. The PDE form in the discussion section is the complete spatial theory.

5 Conclusion

The spatial formation factor C scales primarily with the turnover rate ν as $C \propto \nu^{-0.40}$ ($R^2 = 0.994$), while field diffusion κ has a near-flat effect ($C \propto \kappa^{-0.07}$). The buildup timescale in VCML is governed by the birth-mediated copy-forward propagation mechanism, not by explicit field diffusion. In PDE language, $D_{\text{eff}} \approx D_{\text{copy}}(\nu) \gg \kappa$ for standard parameters, and $D_{\text{copy}} \propto \nu^{0.40}$. This closes the analytical theory of VCML at the mean-field level and identifies the physical origin of the $\sim 9\times$ buildup gap between the ODE and the simulation.

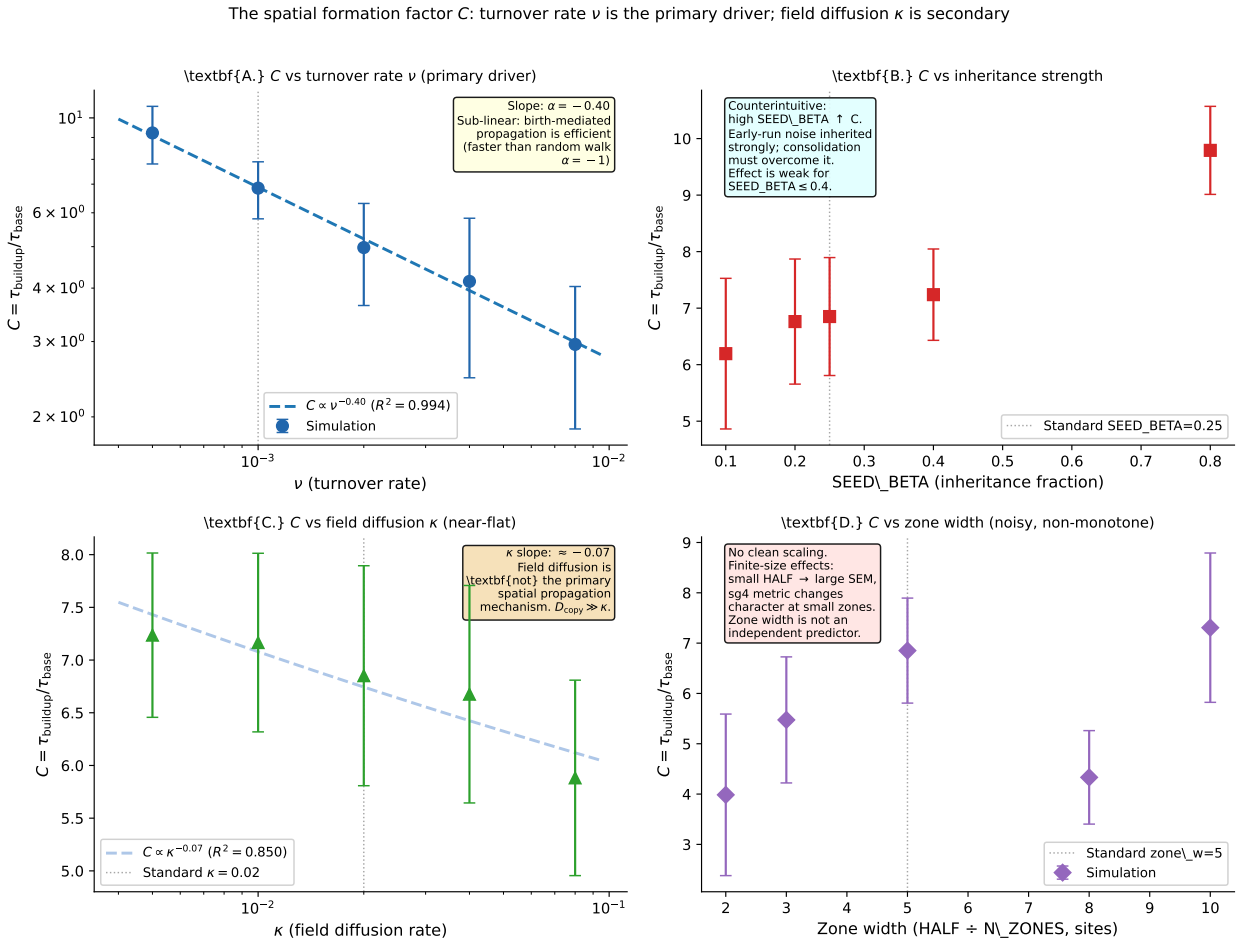

Figure 1: **The spatial formation factor C across four parameter sweeps. A**: C vs turnover rate ν (log-log). Clean power law $C \propto \nu^{-0.40}$ ($R^2 = 0.994$). Turnover rate is the primary driver. **B**: C vs SEED_BETA (inheritance fraction). Weak counterintuitive positive effect: high inheritance slows buildup by seeding newborns with large early-run noise. Effect negligible for SEED_BETA$\leq$ 0.40. **C**: C vs κ (field diffusion). Near-flat ($\alpha = -0.07$). Field diffusion is not the primary spatial propagation mechanism; $D_{\text{copy}} \gg \kappa$. **D**: C vs zone width. Non-monotone, high variance; zone width is confounded by finite-size effects and is not an independent predictor.